

Postfix

Now, let's spend some time on the basics of Postfix configuration, though this is itself a fairly vast topic requiring a lot of knowledge about Internet mail, RFC, etc, which obviously cannot be explored in a single article. In short, Postfix uses two files, `/etc/postfix/main.cf` and `/etc/postfix/master.cf`, for configuring its e-mail service. The commonly edited file is `main.cf`. The default installation of Postfix on OpenBSD will run out-of-the-box, and you can send and receive mail. In case you wish to enable the submission port TCP 587 for sending mail, you can do so by uncommenting (remove the leading hash) the following line in `/etc/postfix/master.cf`:

```
#submission inet n      -       -       -       smtpd
```

After you modify the configuration, to activate it, run a `postfix reload` command. You still have to do some more work in `/etc/postfix/main.cf` to set the local network for relaying, selecting the relay domains and so on. Postfix has built-in controls to guard against becoming an open relay that can be abused by spammers on the Internet, requiring you to set the `mynetworks` parameter to specify from which networks it will accept mail to relay (for example, `mynetworks = 127.0.0.1, 192.168.0.0/16`).

You also need to set the `mydomain`, `relay_domains` and a few other minimal parameters for an excellent standards-compliant full-blown Internet mail server. So the complete configuration looks somewhat like what's shown below:

```
mynetworks = 127.0.0.1, 192.168.0.0/16
myhostname = lfy.com
relay_domains = $mydestination, lfy.com
home_mailbox = Maildir/
```

You can make the most of these things if you already know and handle mail in your day job. You can opt for a 10/8 or 172.16/12 network style in the `mynetworks` parameter, if you have a lot of hosts on your LAN. Always remember that local users can send mail to any destination domain, but outside users can send mails only to those domains specified in the `relay_domains` parameter; be very careful about that. If you test for an open relay configuration, always test from an Internet location; Postfix should reply with a *554 relay access denied* error when you try relaying mail using your mail server, but 'as an outsider (from a network other than the local network)', I emphasise this since this has tripped me up in the past, and I have suffered many hours of agony when spammers hit us. Anyway, only experience can teach you how to use OpenBSD and Postfix to your advantage—no article can do that for you.

Let us now move on to the other topics this article intends to cover.

Web servers

To enable the built-in OpenBSD Apache (there's no need to

install any package), one usually just issues the following line:

```
httpd_flags= in /etc/rc.conf.local
```

...before rebooting (though do not do so yet). That would run a Web server `chroot`-ed to `/var/www`. You can modify files under `/var/www/htdocs`, which is the document root of the Web server. However, for Roundcube, you need to run a non-`chroot` Web server, so use the modified flags line given below:

```
httpd_flags="-u"
```

Now your Web server is insecure, so be careful about exposing it to the Net.

NAS

Next, you may wish to serve files and sometimes also support Windows file-sharing (what is usually meant by NAS), plus also back up and restore in some cases. For running an NFS-mountable server, just modify `/etc/exports` as shown below:

```
/home -mapall=root -network=192.168.1 -mask=255.255.255.0
```

Then, you need the following lines in `/etc/rc.conf.local`:

```
portmap=YES
nfsd_flags=""
```

After a reboot, run `showmount -e` to see if your machine can serve files via NFS.

There are excellent backup and restore utilities that can back up the entire file system. Read the man pages of `dump(8)` and `restore(8)`. The typical usage for `dump` is as shown below:

```
# dump af file.dump /dev/rwd0a
```

For restore, use the following command:

```
# restore rf file.dump
```

It is actually more complicated than that, but you can get started with this. If in doubt, you can always Google to learn more.

I hope this was a good introduction to certain useful things that you can do with OpenBSD. You are even welcome to customise it to suit your unique needs. I'll catch you later with more cool stuff. 

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